

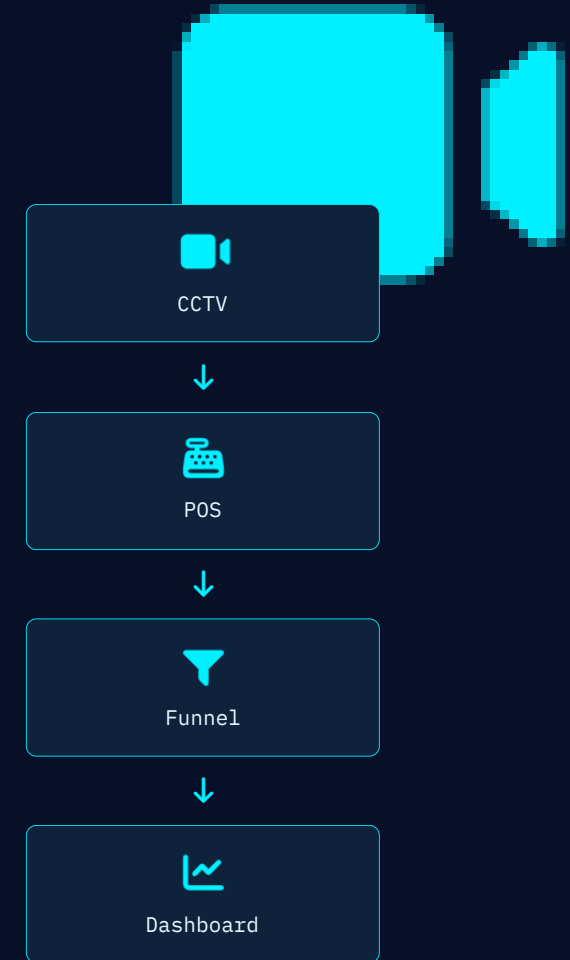
STORE INTELLIGENCE

Store Intelligence: From CCTV + POS to a Live Conversion Funnel

| Suraj Singh

FastAPI · OpenCV · Honest Metrics · Evaluator-Ready in One Command

Backend :8000 · Dashboard :8080 · 5/5 regression tests passing



Retailers know what sold. They don't know who walked away.

- POS systems see only purchasers, missing 50–80% of foot traffic
- CCTV is recorded but rarely fused with transactions
- Most tools inflate conversion by treating buyers as the baseline — rate becomes 100%, hiding the funnel



What would it take to see the full funnel, honestly, in a single pane of glass?

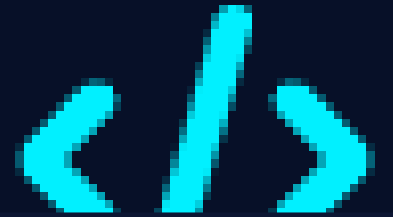
One pipeline. Two signals. One honest funnel.



Auto-discovers inputs by convention. No retraining, no cloud dependency, runs in a single container.

Store Intelligence

Engineering, not magic.



Vision (OpenCV)

OpenCV

```
frame_stride=10
max_frames=600
contours ≥600 px2
tripwire crossing detection
time-budget: 20s/camera
```

Fusion (Python)

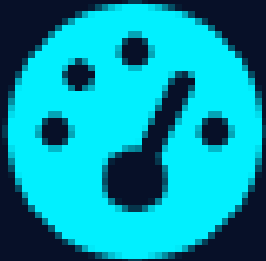
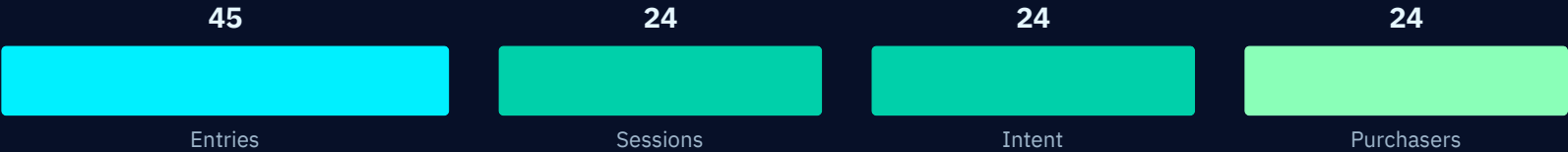
Python

```
session-event lineage graph
transaction reconciliation
anomaly scoring
schema diagnostics
```

What the system reports right now.



Monotonic Funnel



Built for evaluators, not just demos.



Typed API surface

FastAPI

Pydantic models, FastAPI, OpenAPI spec at /docs



Diagnostics endpoints

/diagnostics/schema and /diagnostics/quality expose data-quality flags



Containerized

Docker

docker compose / podman-compose auto-detected, runs in 90 seconds

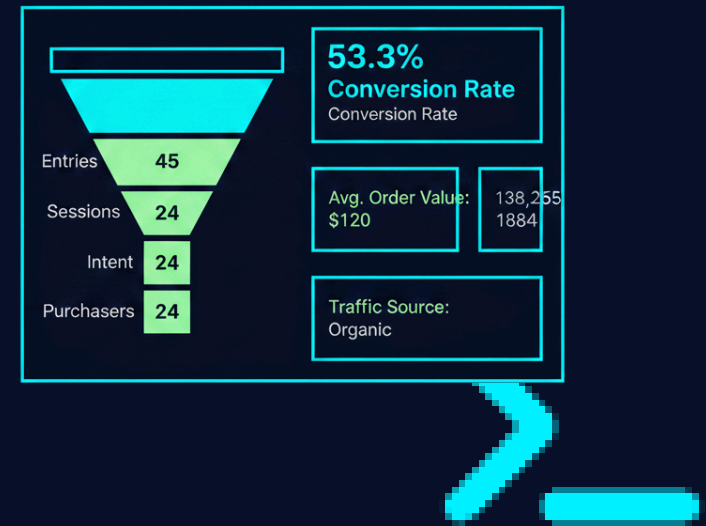


Acceptance harness

smoke + regression + rubric-mapped report (scored 100/100 internally)

Bring your own data. Three steps.

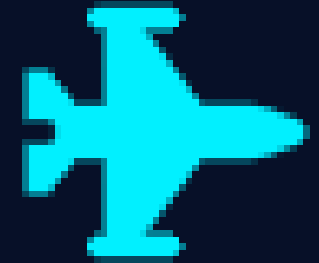
- 1 Drop CCTV .mp4 files into `inputs/cctv/`
- 2 Drop transactions .csv into `inputs/transactions/`
- 3 Run `./scripts/run-stack.ps1 -Build -Detach` [Docker](#)



Zero config edits. Zero file renames. Zero cloud accounts.

From pilot to production.

- Per-zone heatmaps from the same vision pipeline
- Multi-store federation via the existing JSON contract
- Alerting on conversion drops with the anomaly engine already in place



"We didn't build a demo. We built a contract — between cameras, registers, and decisions."

Thank you. Questions welcome.